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Big Data Project Proposal

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Idea

Analyze historical football match data to uncover key performance indicators influencing match outcomes by evaluating and interpreting results under different match conditions, as well as make predictions for future match outcomes.

Dataset

Football matches dataset

Planned approach

EDA

We will analyze how different match statistics correlate to each other and to the match outcomes by plotting graphs of statistics drawn from the data such as:

1. Goals' distributions across leagues and seasons
2. Red cards impact on scoring, possession, and outcome
3. Correlation Heatmaps between shots, fouls, possession, and goals

Descriptive Analysis using association rules

We will use Apriori to find association rules between the data such as:

1. High shot count + home venue → win
2. Red card received + away team → loss

And calculate the associated support, confidence and lift

Descriptive Analysis using clustering

We will use K-Means to segment teams into different clusters such as:

1. Tier segmentation (Strong - Moderate - Weak)
2. Playing-style segmentation (possession-based, counter-attacking, direct play)

Predictive Analysis

Linear regression: To predict goal and foul count

Logistic regression: To predict which team is more probable to win

MapReduce: Apriori will be implemented using MapReduce